

DSA3050A Quiz 1 Submission

Student Name: Tanveer

Student ID: 752

Dataset Source: Brazilian E-Commerce Public Dataset by Olist

Dataset URL: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Dataset Proof



OLIST AND 3
COLLABORATORS · UPDATED
4 YEARS AGO



3932



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Brazilian E-Commerce Public Dataset by Olist

100,000 Orders with product, customer and reviews info

Data Card

Code (691)

Discussion (67)

Suggestions (0)

About Dataset

Usability ⓘ

10.00

Brazilian E-Commerce Public Dataset by Olist

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Dataset Overview

Data Explorer

Version 2 (126.19 MB)

-  olist_customers_dataset.csv
-  olist_geolocation_dataset.csv
-  olist_order_items_dataset.csv
-  olist_order_payments_dataset
-  olist_order_reviews_dataset.c:
-  olist_orders_dataset.csv
-  olist_products_dataset.csv
-  olist_sellers_dataset.csv
-  product_category_name_trans

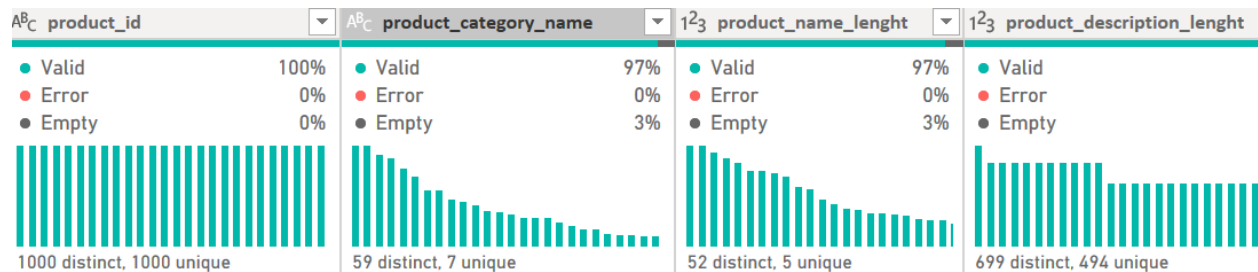
-  olist_customers_dataset
-  olist_order_items_dataset
-  olist_orders_dataset
-  olist_products_dataset
-  product_category_name_translat...
-  olist_order_reviews_dataset

Section A: Power Query (18 Marks)

Q1: Data Profiling + Quality Checks (4 marks)

Data Profile Settings

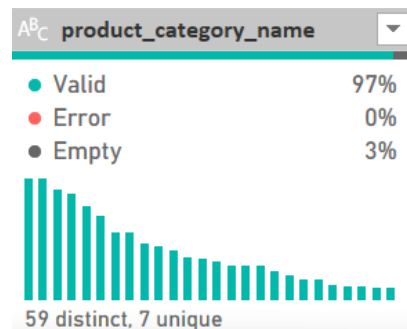
Power Query with Column quality, distribution, and profile turned on (View tab)



Data Quality Issues Found

Issue #1 – Missing values

There are **Missing Values** in the products_dataset in the product_category_name column.



Fix Applied

Replace Values

Replace one value with another in the selected columns.

Value To Find

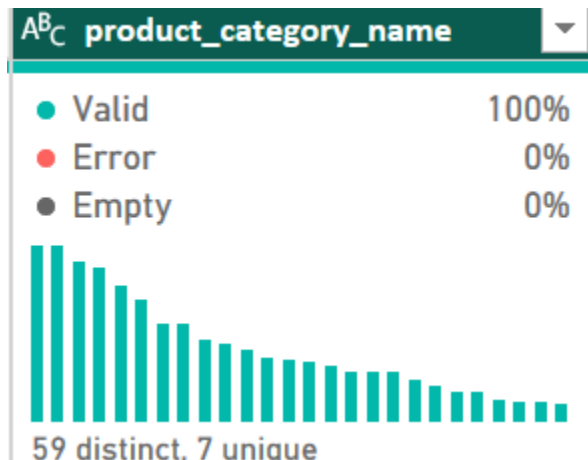
null

Replace With

uncategorized

▸ Advanced options

Issue #1 resolved



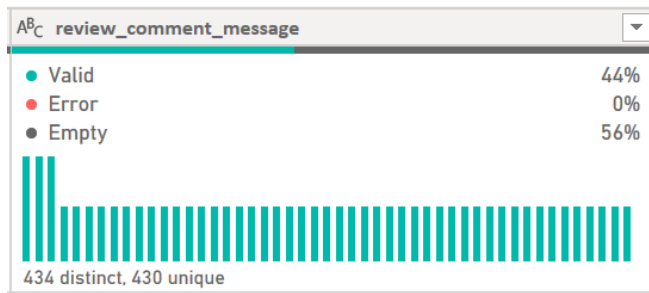
Issue #2 – Missing values

There is a significant proportion of Missing Values in the order_reviews_dataset under the review_comment_message.

- Since there is such a large number missing, deleting would reduce the dataset by 56% which is not a feasible solution.

- I opted to standardize by replacing it with a standard “no comment provided” message.

Issue #2 evidence showing the problem



Fix Applied:

Replace Values

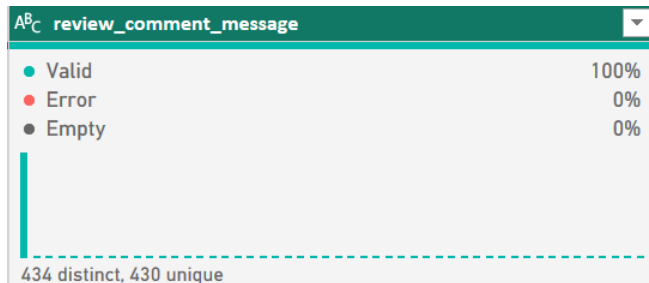
Replace one value with another in the selected columns.

Value To Find

Replace With

▸ Advanced options

Issue #2 resolved



- this solved the empty error

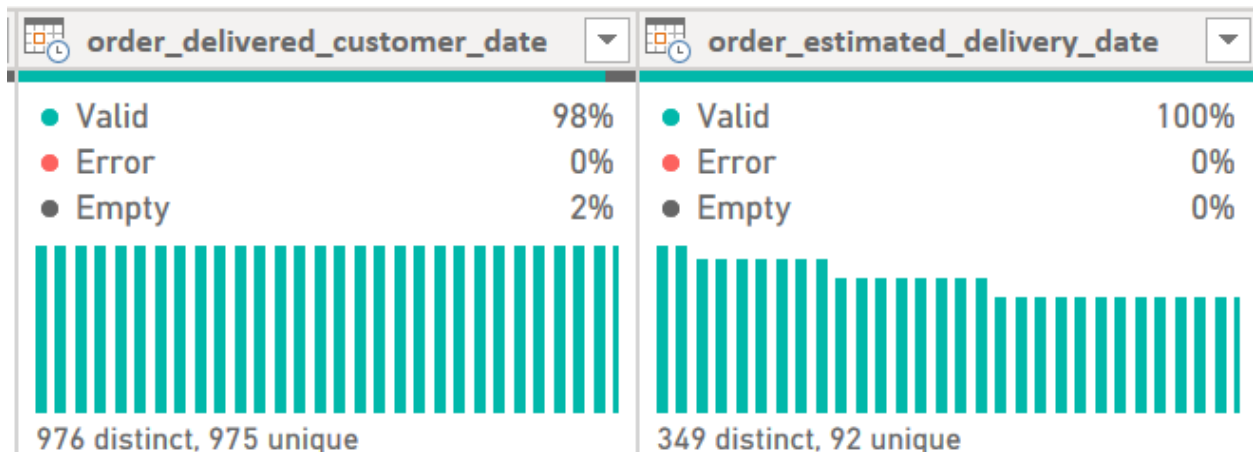
Issue #3 Date Inconsistencies

There are Null Values in the delivery_dates for delivered orders, in the orders_dataset.

My fix:

- add a conditional column called Delivery_Issues
- If order_delivered_customer_date is null and order status is "delivered" then "missing delivery date" , else "OK"

Issue #3 evidence showing the problem



Issue #3 solution Step

Custom Column

Add a column that is computed from the other columns.

New column name

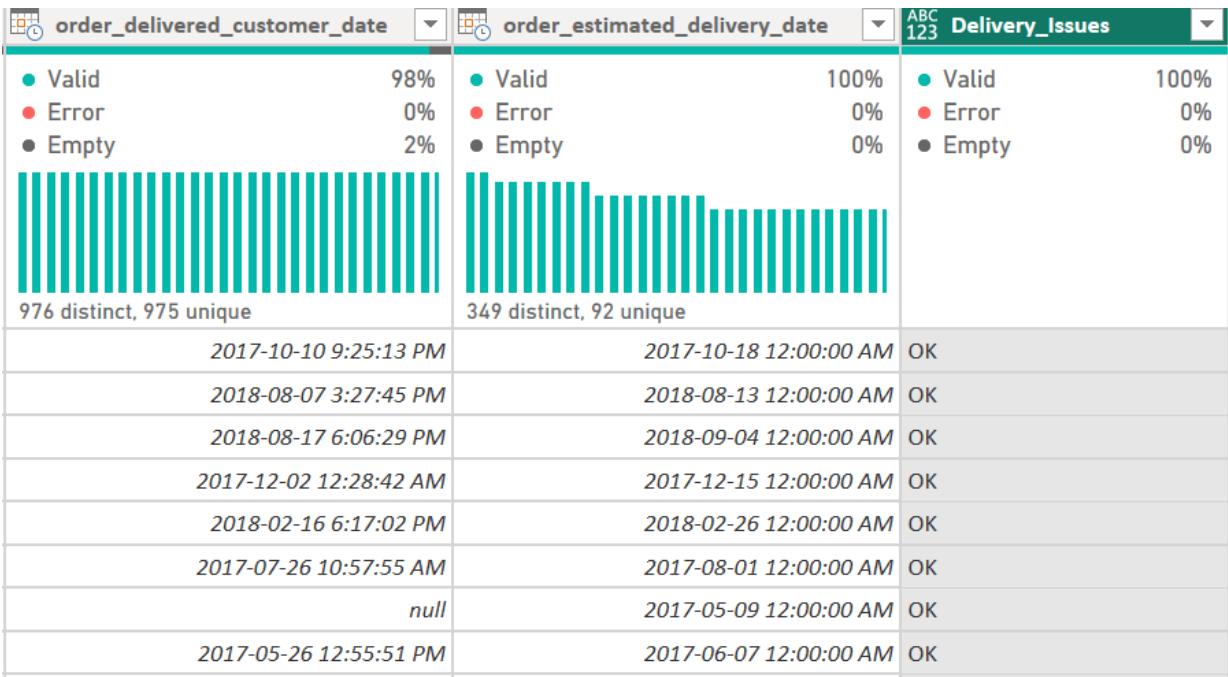
Delivery_Issues

Custom column formula ⓘ

```
= if [order_delivered_customer_date] = null and  
[order_status] = "delivered" then "Missing Delivery Date"  
else "OK"
```

[Learn about Power Query formulas](#)

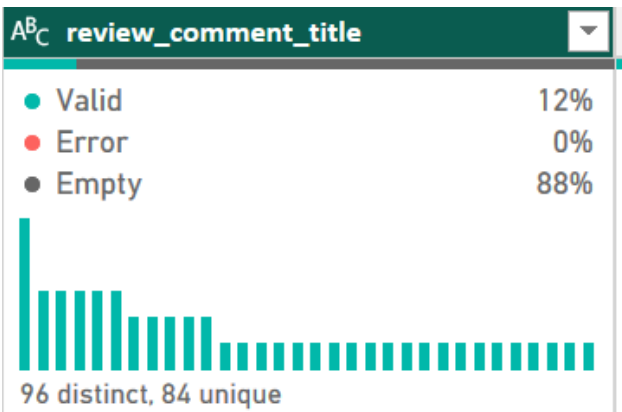
Issue #3 resolved



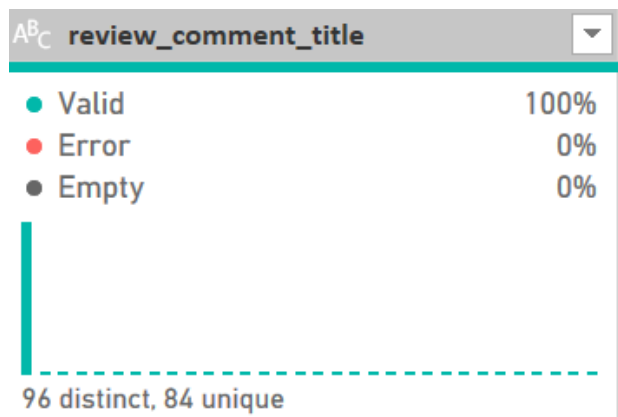
- We can see that even if there are nulls in the delivery date they have in fact been delivered

Issue #4 – Empty Data fields

In the order reviews dataset, the comment titles have a huge amount of empty data points



Ill fix this by replacing with the phrase “no Title” using the same steps as above



Applied Steps Summary for Q1

PROPERTIES

Name

order_reviews_dataset

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Replaced Value
- Replaced Value1
- Trimmed Text
- Cleaned Text
- Lowercased Text
- Capitalized Each Word

PROPERTIES

Name

products_dataset(dimension table)

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Replaced Value
- Replaced Value1

PROPERTIES

Name

orders_dataset

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Added Custom

PROPERTIES

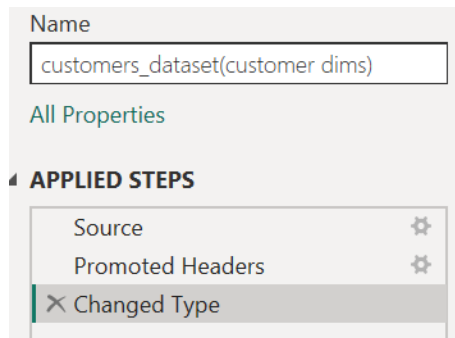
Name

order_items_dataset(fact table)

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type



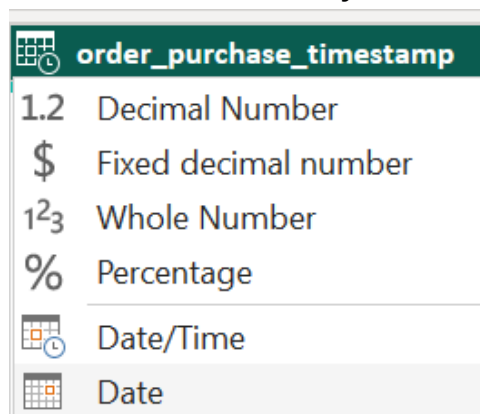
Q2: Data Types + Locale Conversion (4 marks)

Data Type Corrections

Date Column

From orders_dataset

- order_purchase_timestamp
- the data is in date-time format, and we are changing it to date for better time series analysis



- Changed it to date format
- Changed all the timestamps to date for better time series analysis

order_purchase_timestamp	order_approved_at	order_delivered_carrier_date
Valid 100% Error 0% Empty 0% 463 distinct, 193 unique	Valid 100% Error 0% Empty 0% 1000 distinct, 1000 unique	Valid 99% Error 0% Empty 1% 986 distinct, 981 unique
2018-08-08	2018-08-08	2018-08-08
2017-11-18	2017-11-18	2017-11-22
2018-02-13	2018-02-13	2018-02-14
2017-07-09	2017-07-09	2017-07-11
2017-04-11	2017-04-13	null

Whole Number Column

From order_items_dataset

- Order item id is set to decimals but only has whole numbers

1.2 order_item_id

Valid 100%

Error 0%

Empty 0%

5 distinct, 1 unique

1.2 order_item_id

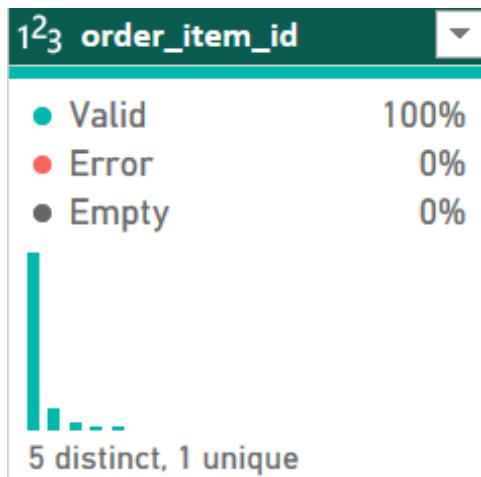
1.2 Decimal Number

\$ Fixed decimal number

123 Whole Number

% Percentage

Fixed:



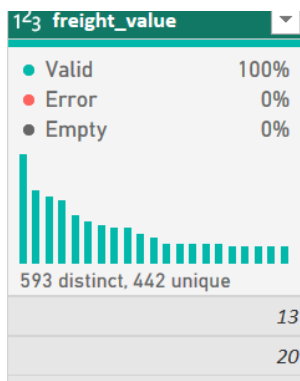
- Changing order_item_id to Whole Number type as quantities cannot be decimals

Decimal/Currency Column

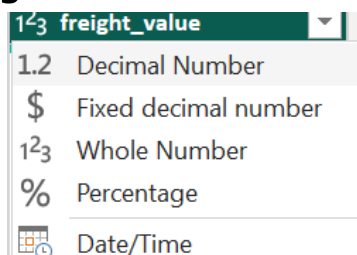
From the order_items_dataset

- Changing freight value from whole number to decimal place

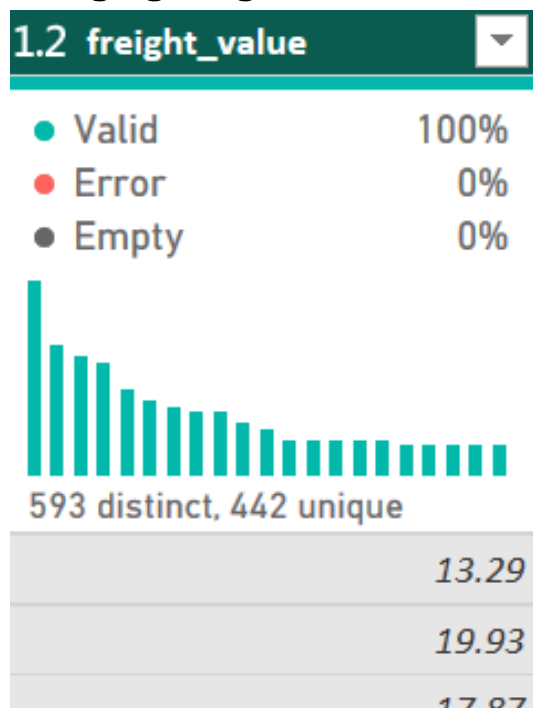
Original:



Changed:



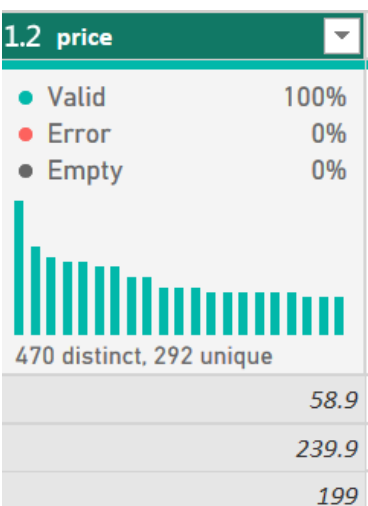
Changing freight value to Decimal type (look at top left of image)



Locale Conversion

Column Selected for Locale Conversion: price

Before Conversion:



Before locale conversion - numbers incorrectly formatted, such as no set decimals

Locale Settings Used:

Change Type with Locale

Change the data type and select the locale of origin.

Data Type

Fixed decimal number

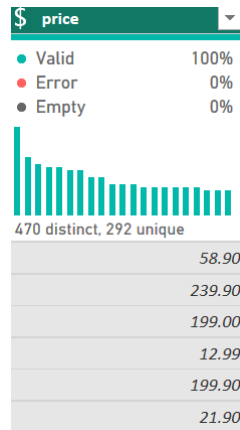
Locale

English (Canada)

i Sample input values:

2,100.50

-1.50



- Set the locale to English (Canada), with data type fixed decimal places as is better for currency

Applied Steps for Q2

Applied Steps showing data type changes and locale conversion:

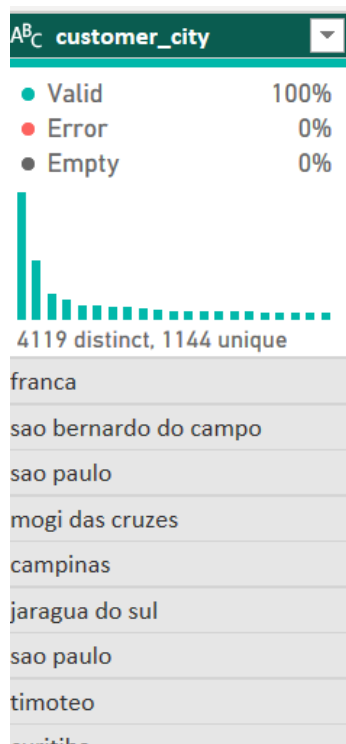
Name	order_reviews_dataset	products_dataset(dimension table)
order_items_dataset(fact table)	order_reviews_dataset	products_dataset(dimension table)
All Properties	All Properties	All Properties
APPLIED STEPS	APPLIED STEPS	APPLIED STEPS
Source	Source	Source
Promoted Headers	Promoted Headers	Promoted Headers
Changed Type	Changed Type	Changed Type
Replaced Value	Replaced Value	Replaced Value
Replaced Value1	Replaced Value1	Replaced Value1
Trimmed Text	Trimmed Text	Trimmed Text
Cleaned Text	Cleaned Text	Cleaned Text
Lowercased Text	Lowercased Text	Lowercased Text
Capitalized Each Word	Capitalized Each Word	Capitalized Each Word
Changed Type with Locale	Changed Type1	Changed Type1

Q3: Text Standardization (4 marks)

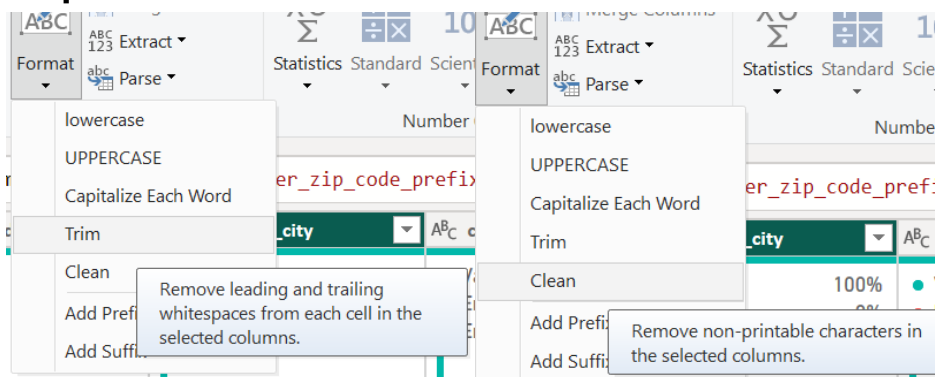
Column Selected for Standardization:

- From customer_dataset
 - Customer city
 - Data is haphazard and doesn't follow correct format for cities
 -

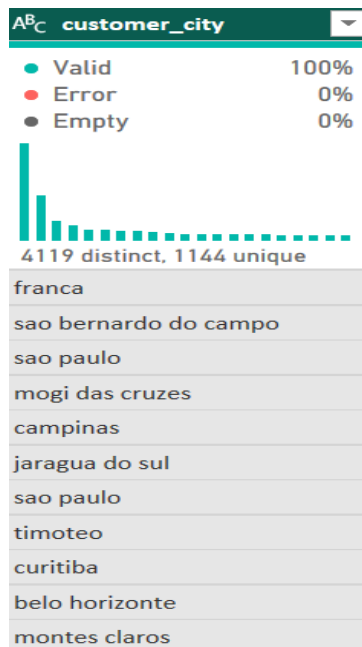
Before Standardization:



Step 1: Trim + Clean



CLEANED:

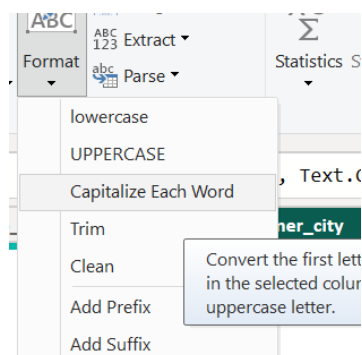


After applying Trim and Clean:

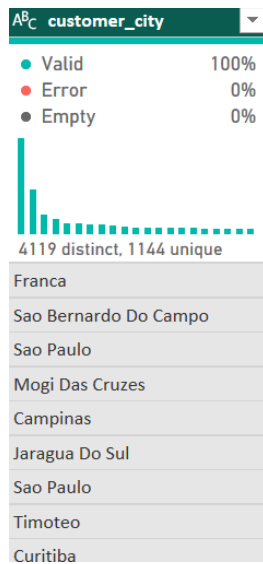
- spaces and non-printable characters removed
- Removed leading/trailing spaces and non-printable characters

Step 2: Standardize Casing

Action: Transform → Format → Proper Case



After Casing



What changed:

- Converted to Proper Case where each word starts with capital letter

customers_dataset(customer dims)

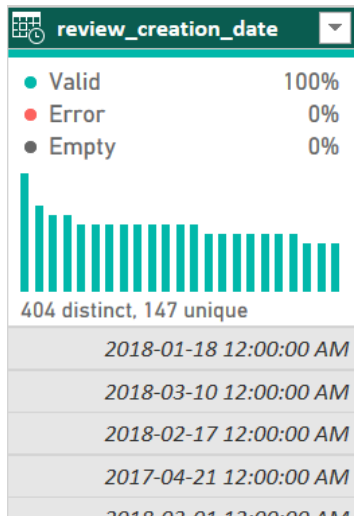
All Properties

APPLIED STEPS

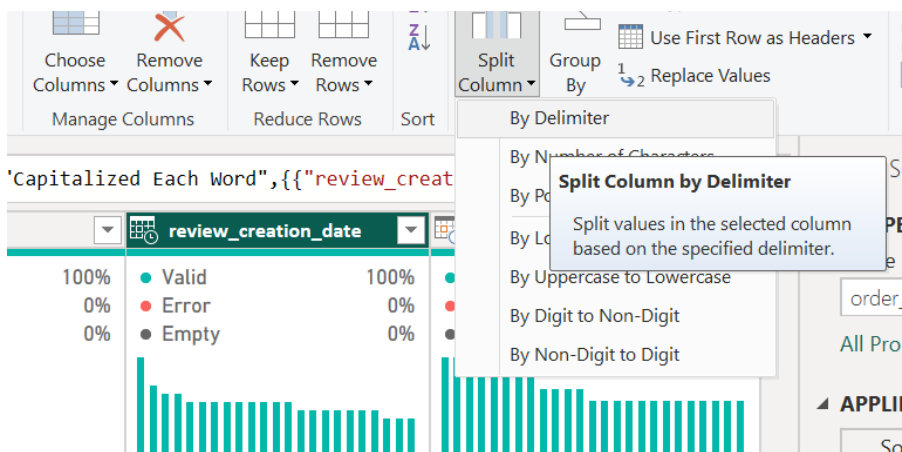
- Source
- Promoted Headers
- Changed Type
- Trimmed Text
- Cleaned Text
- Lowercased Text
- Capitalized Each Word**

Step 3: Split Column

From order reviews table, review_creation_date column:



I split this into date and time



Split Column by Delimiter

Specify the delimiter used to split the text column.

Select or enter delimiter

Space

Split at

☒ Left-most delimiter

☐ Right-most delimiter

☐ Each occurrence of the delimiter

Advanced options

Split into

☒ Columns

☐ Rows

Quote Character

None

☐ Split using special characters

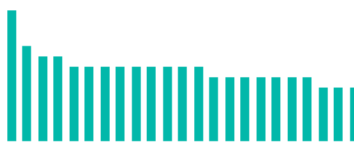
Insert special character

OK

Cancel

Successful split:

- Column split into 2 new columns using space delimiter

review_creation_date	review_creation_time
Valid 100% Error 0% Empty 0%  404 distinct, 147 unique	Valid 100% Error 0% Empty 0%  2 distinct, 0 unique
2018-01-18	12:00:00 AM
2018-03-10	12:00:00 AM
2018-02-17	12:00:00 AM
2017-04-21	12:00:00 AM
2018-03-01	12:00:00 AM

Step 4: Rename Columns

Renaming:

review_creation_date.2

Valid
Error
Empty

2 distinct, 0 unique

review answer timestamp

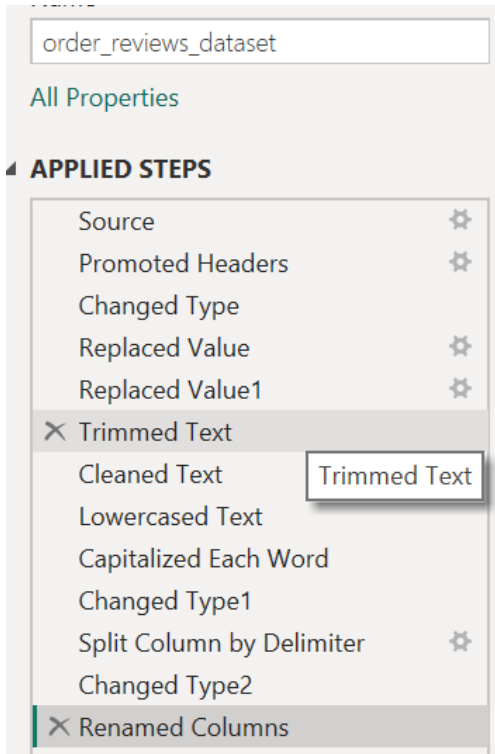
Copy
Remove
Remove Other Columns
Duplicate Column
Add Column From Examples...
Remove Duplicates
Remove Errors
Change Type
Transform
Replace Values...
Replace Errors...
Group By...
Fill
Unpivot Columns
Unpivot Other Columns
Unpivot Only Selected Columns
Rename...

Final text columns after renaming:

review_creation_date review_creation_time

Complete Applied Steps for Q3:

All text transformation steps visible in Applied Steps pane:



Q4: Conditional + Custom Columns (4 marks)

Conditional Column

Column Purpose:

Categorize price into price tiers

Conditional Logic:

If price $\geq$ 200 then "Premium"

Else if price $\geq$ 100 then "High"

Else if price $\geq$ 50 then "Medium"

Else "Budget"

New column name: Price Tier

Conditional column configuration window:

Custom Column

Add a column that is computed from the other columns.

New column name

Price_Tier

Custom column formula ⓘ

```
= if [price] >= 200 then "Premium"
else if [price] >= 100 then "High"
else if [price] >= 50 then "Medium"
else "Budget"
```

Available columns

order_id

order_item_id

product_id

seller_id

shipping_limit_date

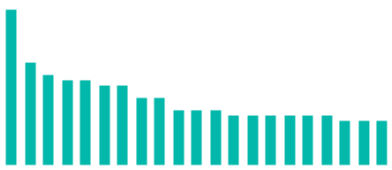
price

freight_value

<< Insert

[Learn about Power Query formulas](#)

Result:

\$ price		ABC 123 Price_Tier	1.2
Valid100% Error0% Empty0%  470 distinct, 292 unique		Valid100% Error0% Empty0%  59%	
9	58.90	Medium	
3	239.90	Premium	
8	199.00	High	
5	12.99	Budget	
3	199.90	High	
3	21.90	Budget	
4	19.90	Budget	
0	810.00	Premium	
6	145.95	High	

Custom Column (with M Formula)

Column Purpose: revenue per item

M Formula Used:

New column name

Revenue

Custom column formula ⓘ

```
= [order_item_id]*[price]
```

ABC 123 Revenue	
Valid	- %
Error	100%
Empty	- %

58.9
239.9
199
12.99
199.9
21.9
19.9

Column Purpose: Freight Percentage
New column name: Freight_Percentage

Formula:
[freight_value]/[price]*100

New column name

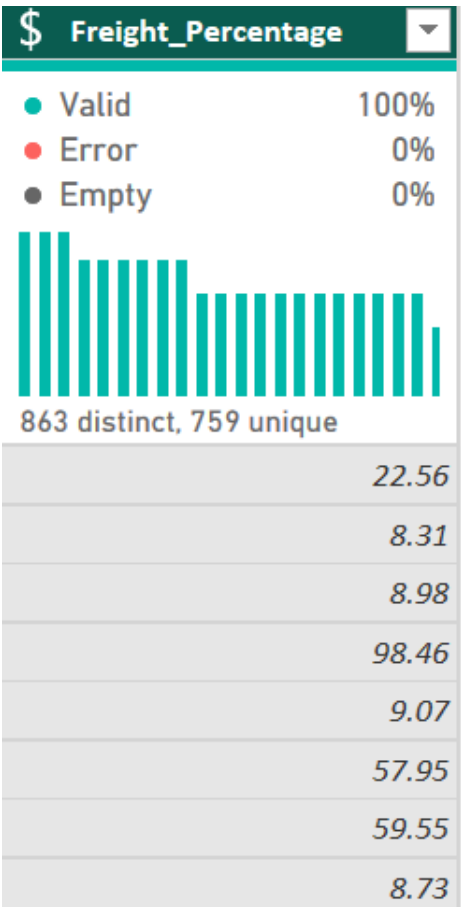
Freight_Percentage

Custom column formula ⓘ

= ([freight_value] / [price]) * 100

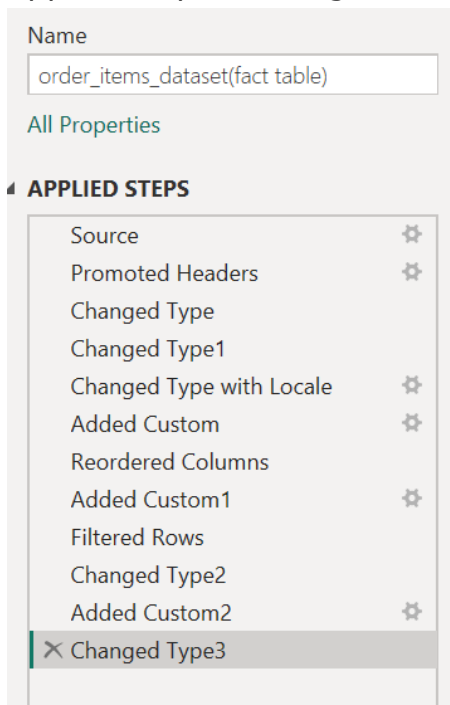
Average of price

Formula Explanation:
The value of the freight is divided by price to get the freight price ratio and then again multiplied by 100 to get the percentage



Applied Steps for Q4:

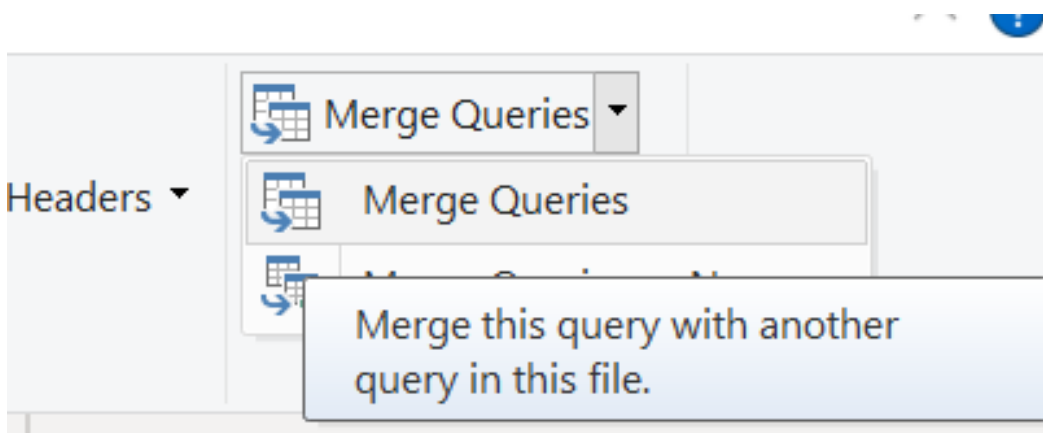
Applied Steps showing conditional and custom column additions:



Q5: Grouping OR Merge (2 marks)

Chosen Operation: Merge Queries

Merge Queries:



Merge

Select a table and matching columns to create a merged table.

products_dataset(dimension table)

product_id	product_category_name	product_name_lenght	product_d
1e9e8ef04dbcff4541ed26657ea517e5	Perfumaria	40	
3aa071139cb16b67ca9e5dea641aaa2f	Artes	44	
96bd76ec8810374ed1b65e291975717f	Esporte_Lazer	46	
cef67bcfe19066a932b7673e239eb23d	Bebes	27	
...	...	...	...

product_category_name_translation

product_category_name	product_category_name_english
Beleza_Saude	Health_Beauty
Informatica_Acessorios	Computers_Accessories
Automotivo	Auto
Cama_Mesa_Banho	Bed_Bath_Table
Moveis_Decoracao	Furniture_Decor

Join Kind

Left Outer (all from first, matching from second)

Expanding Product Category:

Expand product_category_na...

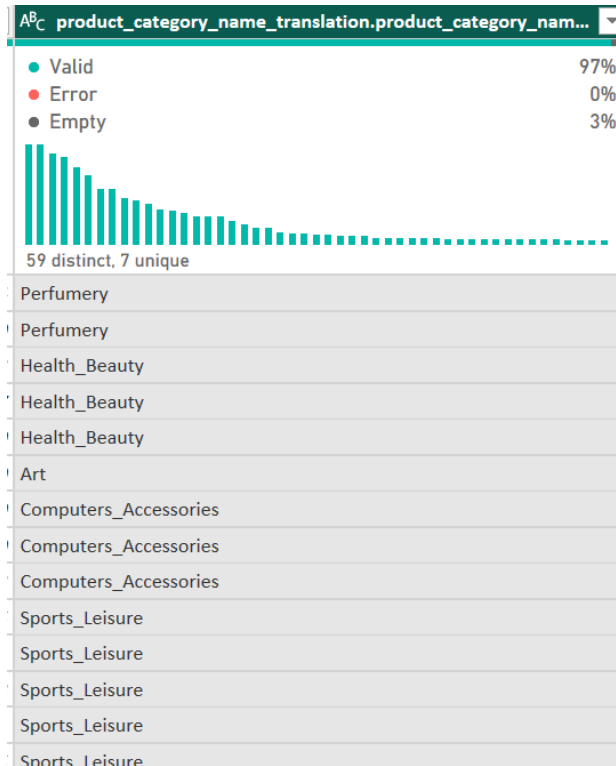
Select the columns to expand.



- ☒ (Select All Columns)
- ☐ product_category_name
- ☒ product_category_name_english

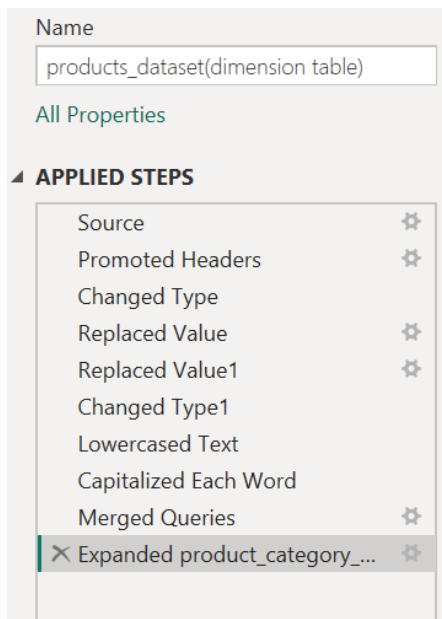
Results:

Main table with additional columns expanded from lookup table, and translated to English from Portuguese



Applied Steps:

Applied Steps showing Merge and Expand operations:



Section B: Data Modeling (12 Marks)

Q6: Star Schema Identification (3 marks)

Table Classification

- **Order_items_dataset** - FACT Table
 - Contains measures, foreign keys, transactional data
- **Orders_dataset** - DIMENSION Tables
 - Order attributes, dates, status
- **Products_dataset** - DIMENSION Table
 - Product descriptions, categories
- **Customer_dataset** - DIMENSION Table
 - Customer location data
- **Order_reviews_dataset** - FACT Table
 - Contains ratings
- **Product_category_name_translation** - DIMENSION Table
 - Lookup table for categories

Detailed Justifications:

The order_items_dataset is a FACT table because:

- It contains numerical measures that can be aggregated: price and freight_value
- It has foreign keys linking to dimension tables: product_id, order_id
- It represents business events: each row is one item in an order
- With ~112,000 rows, it's the largest table and central to analysis

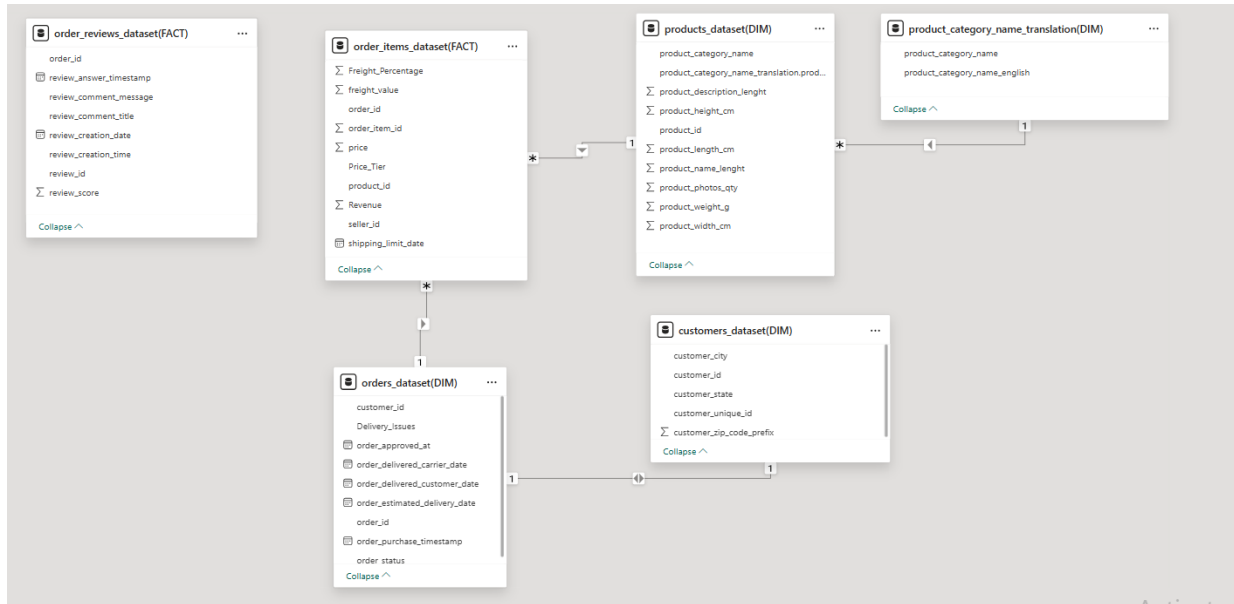
The Products_dataset is a DIMENSION table because:

- It contains descriptive attributes about products: category name, product details
- Each product appears only once (unique product_id)
- It provides context for analyzing facts: "which categories sell best?"
- It contains no numerical measures to aggregate

Q7: Relationships (5 marks)

Model View Screenshot:

Power BI Model view showing all tables and relationships



Relationship 1: Orders -> order_items

From table

order_items_dataset(FACT)

freight_value	order_id	order_item_id	Price_Tier	product_id	Revenue	seller_id
15.1	00c8be06a80...	1	Budget	5d66715cc92...	19.9	1286394
15.1	02605457086...	1	Budget	fed5c40c27e1...	19.9	1286394
15.1	044cd7ffb6a4...	1	Budget	62c2b9bd445...	19.9	002100f

To table

orders_dataset(DIM)

customer_id	Delivery_Issues	order_approv...	order_deliver...	order_deliver...	order_estimat...	order_id
d42ee2c0427...	OK	November-24...	November-27...	December-19...	December-27...	52e5445
04e348e604e...	OK	November-24...	November-27...	December-12...	December-15...	30c26d1
e630c8a8b54...	OK	November-24...	November-27...	November-29...	December-12...	c000556

Cardinality

Many to one (*:1)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

- **Cardinality:** One-to-many
- **Cross filter direction:** Single
- **Active:** Yes

Why this is correct:

One order can contain multiple items. This relationship allows us to see all items that belong to each order. When we filter by order date, it correctly filters the items in those orders.

Note that in POWER BI the many is always the FACT TABLE

Relationship 2 : Order_items → Products_dataset

From table

order_items_dataset(FACT) ▼

freight_value	order_id	order_item_id	Price_Tier	product_id	Revenue	seller_id
15.1	00c8be06a80...	1	Budget	5d66715cc92...	19.9	1286394
15.1	02605457086...	1	Budget	fed5c40c27e1...	19.9	1286394
15.1	044cd7ffb6a4...	1	Budget	62c2b9bd445...	19.9	002100f

To table

products_dataset(DIM) ▼

product_cate...	product_cate...	product_desc...	product_heig...	product_id	product_lengt...	product_n
Fashion_Bolsa...	Fashion_Bags...	770	2	5d2754abdc8...	16	53
Fashion_Bolsa...	Fashion_Bags...	731	2	49feeab25d6...	16	48
Fashion_Bolsa...	Fashion_Bags...	83	2	8f4735c09292...	16	60

Cardinality

Many to one (*:1) ▼

Cross-filter direction

Single ▼

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Save

Cancel

- **Cardinality:** One-to-many
- **Cross filter direction:** Single
- **Active:** Yes

Why this is correct:

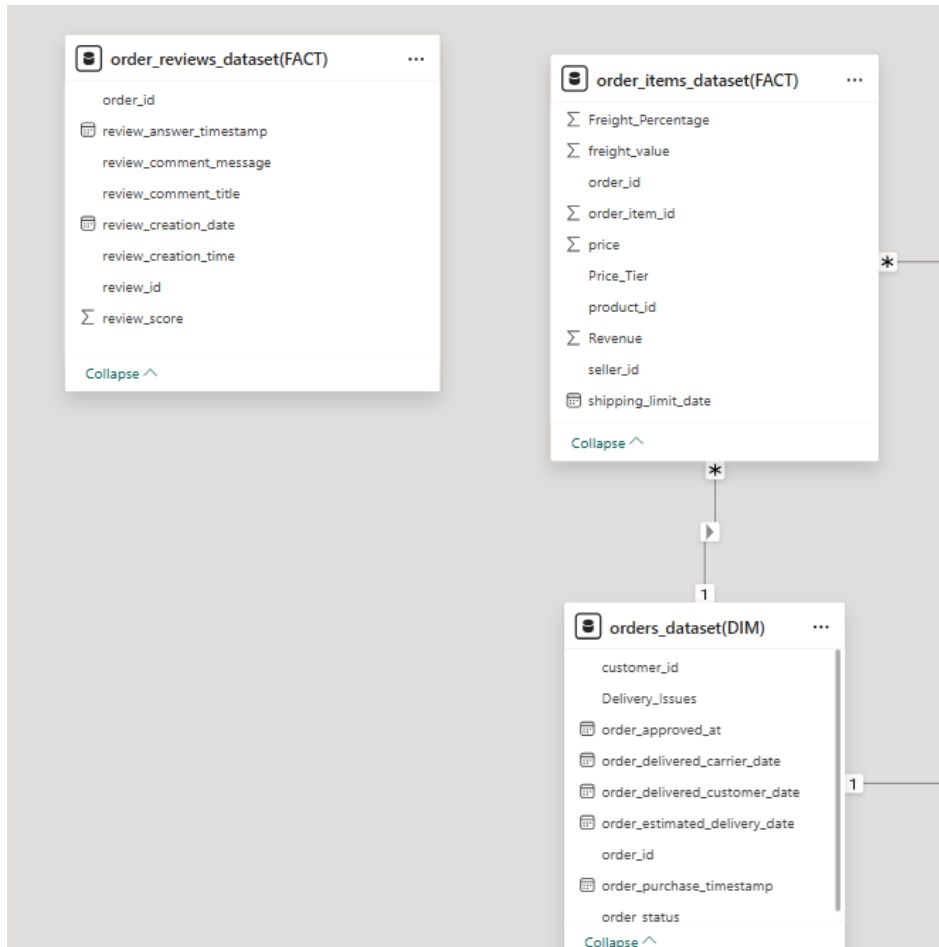
One product can appear in multiple order items across different orders. This lets us analyze sales by product category. Filtering by product category correctly shows all order items for products in that category.

Note that in POWER BI the many is always the FACT TABLE

Modeling Mistake Identification:

Mistake Identified:

Order Reviews has no relationship



Impact on Totals:

Without a relationship between reviews and orders, you cannot analyze review ratings by order date, customer location, or product category. If you try to show average rating per month, it will show the same average for all months because the filter won't propagate to reviews.

How to Fix:

Create a relationship from olist_order_reviews_dataset[order_id] to olist_orders_dataset[order_id] with many-to-one cardinality.

New relationship

Select tables and columns that are related.

From table

order_reviews_dataset(FACT) ▼

order_id	review_answe...	review_comm...	review_comm...	review_creati...	review_creati...	review_id
1aa3c5dc3d9...	August-07-18	No Comment...	no title	August-01-18	12:00:00 AM	c933a19
47c06c1588f9...	August-03-18	No Comment...	no title	August-01-18	12:00:00 AM	32fcd02
f03b33cf5e0a...	August-02-18	No Comment...	no title	August-01-18	12:00:00 AM	a415187

Delete and recreate relationship to select a different table and column

To table

orders_dataset(DIM) ▼

customer_id	Delivery_Issues	order_approv...	order_deliver...	order_deliver...	order_estimat...	order_id
d42ee2c0427...	OK	November-24...	November-27...	December-19...	December-27...	52e5445
04e348e604e...	OK	November-24...	November-27...	December-12...	December-15...	30c26d1
e630c8a8b54...	OK	November-24...	November-27...	November-29...	December-12...	c000556

Cardinality

Many to one (*:1) ▼

Cross-filter direction

Single ▼

☒ Make this relationship active

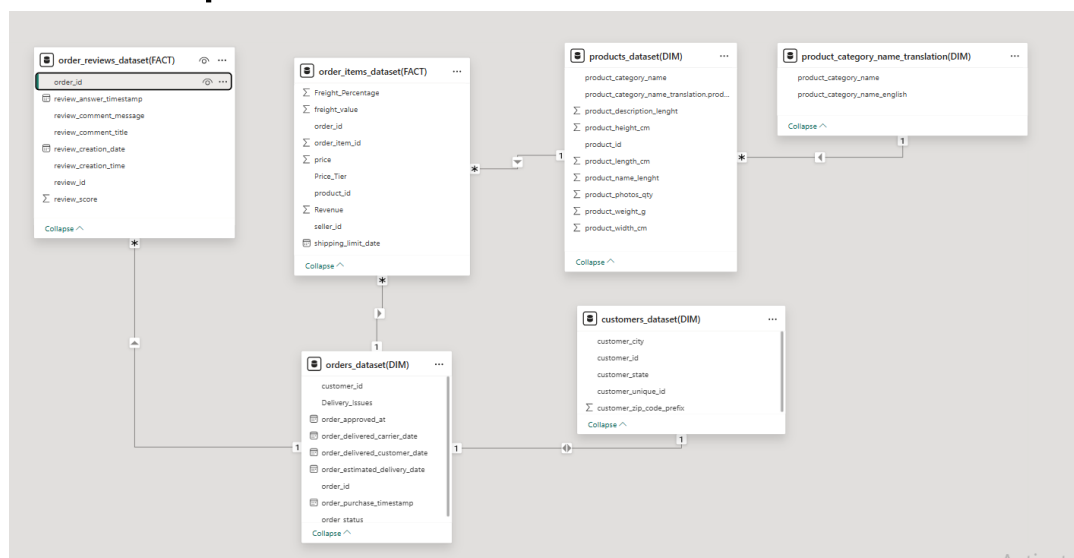
☐ Apply security filter in both directions

☐ Assume referential integrity

Save

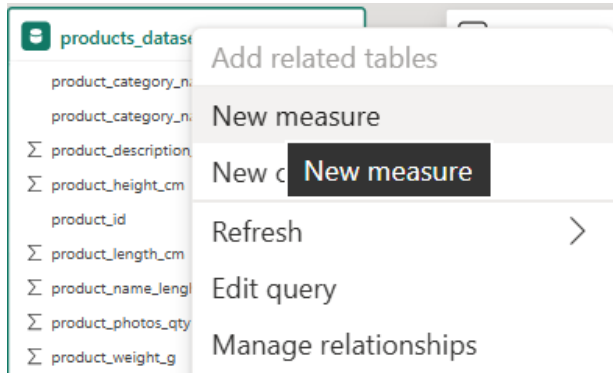
Cancel

Relationship after correction:



Q8: Key Uniqueness Verification (2 marks)

Dimension Table Selected: Products_dataset



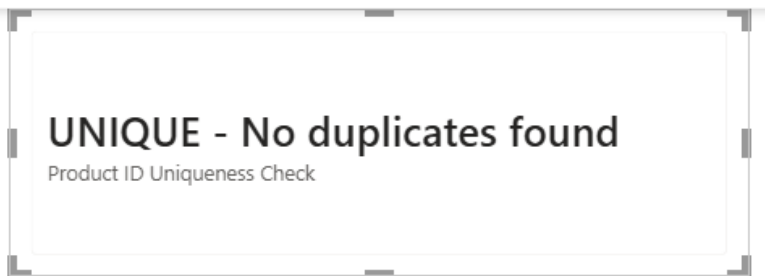
Key Column: ProductID

Verification Method

I used the DAX method as I am used to coding

```
Product ID Uniqueness Check =  
IF(  
    COUNTROWS('products_dataset(DIM)') = DISTINCTCOUNT('products_dataset(DIM)'[product_id]),  
    "UNIQUE - No duplicates found",  
    "DUPLICATES FOUND - Check data"  
)
```

Then I used a card to show output in report view:



Q9: Date Table Decision (2 marks)

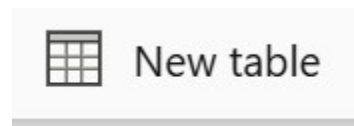
Does a date table need to exist?

My Answer: YES

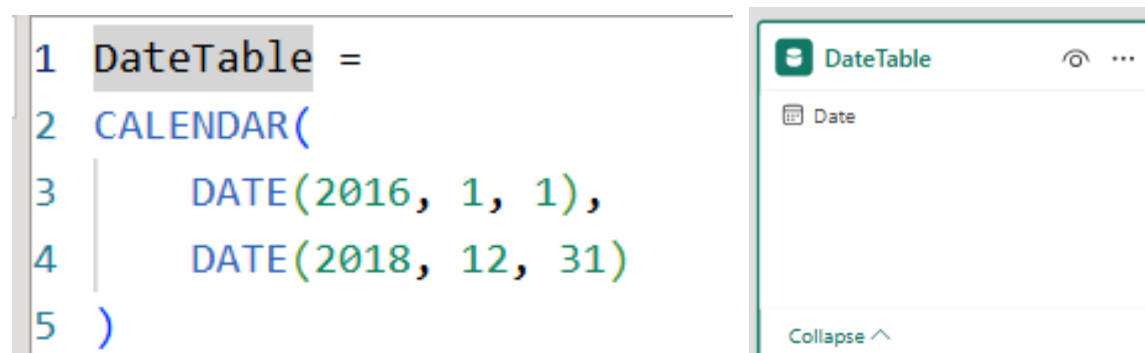
REASONS:

- 1. Time Intelligence Functions:** A separate date table enables DAX functions like TOTALYTD, SAMEPERIODLASTYEAR, and DATEADD which don't work correctly without a proper date table.
- 2. Continuous Timeline:** A date table provides all dates even when no orders occurred, ensuring time series visuals show complete periods without gaps, and allows consistent filtering across multiple fact tables with different date ranges.

First I created a New Table in Model View:



Then I set a time-frame that the table would hold:



Creating columns:

```
1 Year = YEAR(DateTable[Date])
```

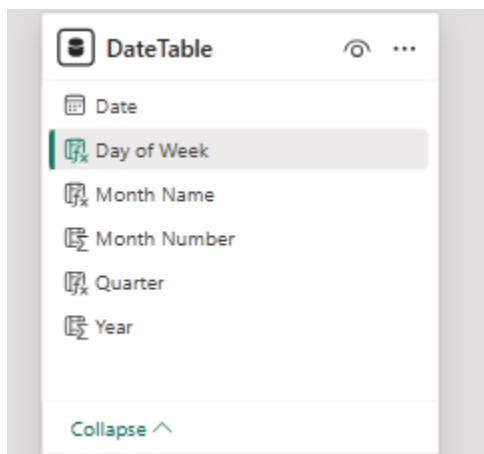
```
1 Month Number = MONTH(DateTable[Date])
```

```
1 Month Name = FORMAT(DateTable[Date], "MMMM")
```

```
1 Quarter = "Q" & QUARTER(DateTable[Date])
```

```
1 Day of Week = FORMAT(DateTable[Date], "dddd")
```

Date Table:



Marked as Date Table:

Mark as a date table

To enable the creation of date-related visuals, tables :

Keep in mind any built-in date tables that are already referring to them may break. [Learn more](#)

Mark as a date table

☒ On

Choose a date column

Date

✔ Validated successfully

Relationship to orders:

Edit relationship

Select tables and columns that are related.

From table

orders_dataset(DIM)

pprov...	order_deliver...	order_deliver...	order_estimat...	order_id	order_purcha...	order_status
ber-24...	November-27...	December-19...	December-27...	52e5445828d...	November-24...	delivered
ber-24...	November-27...	December-12...	December-15...	30c26d11b6d...	November-24...	delivered
ber-24...	November-27...	November-29...	December-12...	c000556923c...	November-24...	delivered

To table

DateTable

Date	Day of Week	Month Name	Month Number	Quarter	Year
01/01/2016 1...	Friday	January	1	Q1	2016
02/01/2016 1...	Saturday	January	1	Q1	2016
03/01/2016 1...	Sunday	January	1	Q1	2016

Cardinality

Many to one (*:1)

Cross-filter direction

Single



Make this relationship active



Apply security filter in both directions

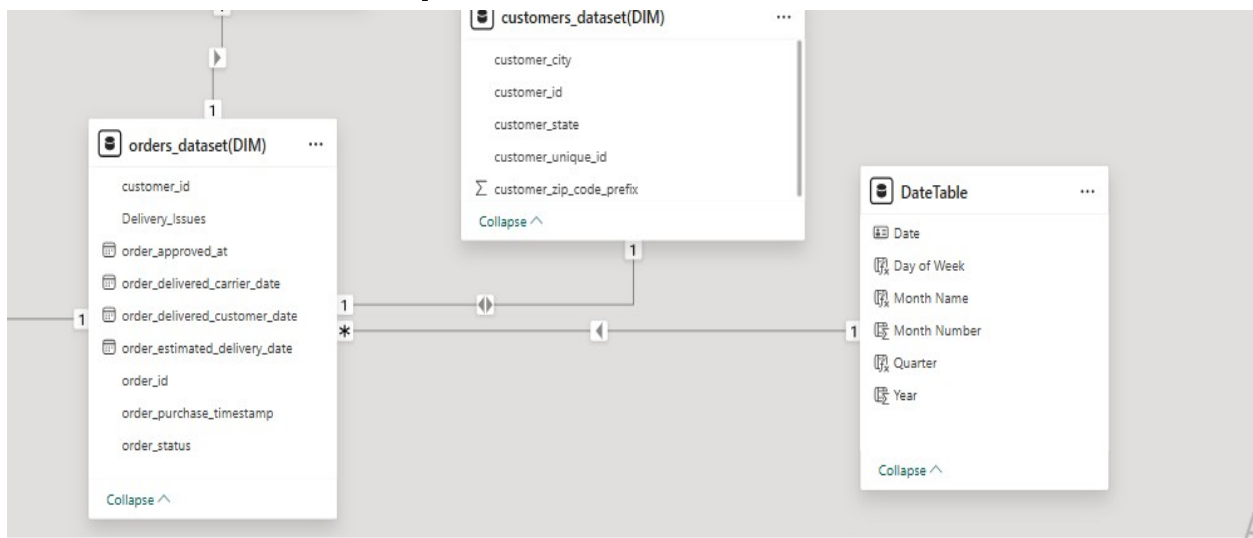


Assume referential integrity

Save

Cancel

Date table Relationship:



Bonus: Small Data Dictionary (+2 marks)

DATA DISCTIONARY

Column	Data Type	Key/Attribute	Meaning	Table
customer_city	Text	Attribute	City of customer	customers_dataset
customer_id	Text	FK	Foreign key to customer	orders_dataset
customer_unique_id	Text	PK	Unique customer identifier	customers_dataset
order_id	Text	FK	Foreign key to order	order_items_dataset
order_id	Text	PK	Unique order identifier	orders_dataset
price	Decimal	Measure	Item price in Brazilian Real	order_items_dataset
product_id	Text	FK	Foreign key to product	order_items_dataset
product_id	Text	PK	Unique product identifier	products_dataset

Simple report:

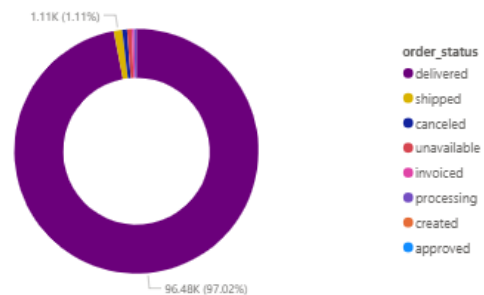
UNIQUE - No duplicates found

Product ID Uniqueness Check

DATA DISCTIONARY

Column	Data Type	Key/Attribute	Meaning	Table
customer_city	Text	Attribute	City of customer	customers_dataset
customer_id	Text	FK	Foreign key to customer	orders_dataset
customer_unique_id	Text	PK	Unique customer identifier	customers_dataset
order_id	Text	FK	Foreign key to order	order_items_dataset
order_id	Text	PK	Unique order identifier	orders_dataset
price	Decimal	Measure	Item price in Brazilian Real	order_items_dataset
product_id	Text	FK	Foreign key to product	order_items_dataset
product_id	Text	PK	Unique product identifier	products_dataset

Count of order_id by order_status



Count of order_purchase_timestamp by Date



Top Selling Items

